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Lab01 .pdf

Helpful Unhelpful

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lab01

June 15, 2023

[12]:
```
import otter
grader = otter.Notebook()
```

1 Lab 1: Expressions

Welcome to Math 211: *Foundations of Data Science*! Each week you wil
like this one. You can't learn technical subjects without hands-or
part of the course.

Before we get started, there are some administrative details.

The weekly lab session has two components: questions and di
about recent material, and a lab assignment (like this one!) that
and inferential concepts. These lab assignments are a required p
on Monday mornings.

Collaborating on labs is more than okay - it's encouraged! You s
than a few minutes on questions in labs, so ask a neighbor or
things is beneficial, too - the best way to solidify your knowle
Please don't just share answers, though.

You can read more of the course policies

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book, but not right now. Instead, let's get started!

2 1. Jupyter notebooks

This webpage is called a Jupyter notebook. A notebook is a plac
results, and also to write text.

1

2.1 1.1. Text cells

In a notebook, each rectangle containing text or code is called *cell*.

Text cells (like this one) can be edited by double-clicking on the
mat called Markdown to add formatting and section headings. You
but you might want to.

After you edit a text cell, click the "run cell" button at the to
shift + return to confirm any changes. (Try not to delete the in

Question 1.1.1. This paragraph is in its own text cell. Try editing i
the last sentence in the paragraph, and then click the "run cel
return. This sentence, for example, should be deleted. So shou

2.2 1.2. Code cells

Other cells contain code in the Python 3 language. Running a c
it contains.

To run the code in a code cell, first click on that cell to activate
green or blue rectangle. Next, either press | or hold shift + return

Try running this cell:

[13]:
```
print("Hello, World!")
```

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they appear. Every print expression prints a line. Run the next c
output.

[15]:
```
print("First this line,")
print("then the whole ")
print("and then this one.")
```

```
First this line,
then the whole ,
and then this one.
```

Question 1.2.1. Change the cell above so that it prints out:

```
First this line,
then the whole ,
and then this one.
```

2

2.3 1.3. Writing Jupyter notebooks

You can use Jupyter notebooks for your own projects or docur
notebook, you'll need to create your own cells for text and cod

To add a cell, click the + button in the menu bar. It'll start out
to a code cell by clicking inside it so it's highlighted, clicking th
() button in the menu bar, and choosing "Code".

Question 1.3.1. Add a code cell below this one. Write code in it

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weeks and gain reasonable proficiency with the language in a
you're proficient in a natural language, you can understand a n
small mistakes. A computer running Python code is not smart e

Whenever you write code, you'll make mistakes. When you run
will sometimes produce error messages to tell you what you di

Errors are okay; even experienced programmers make many er
just have to find the source of the problem, fix it, and move on.

We have made an error in the next cell. Run it and see what ha

[17]:
```
print("This line is missing something."
```

```
  File "/tmp/ipykernel_53/1867411798.py"line 1
    print("This line is missing something."
                                           ^
SyntaxError: unexpected EOF while parsing
```

[ ]:
```
print("This line is missing something.')
```

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The kernel is a program that executes the code inside your no
the top right of your window, you can see a circle that indicat
circle is empty (), the kernel is idle and ready to execute the
kernel is busy running some code.

Next to every code cell, you'll see some text that says `In [ ]:`. Before
see `In [ ]`. When the cell is running, you'll see `In [*]`. If you see an
that doesn't go away, it's likely that the code inside the cell is ta
be a good time to interrupt the kernel (discussed below). Whe
see a number inside the brackets, like `In [10]`. The number corresp
you run the cells; so, the first cell you run will show a 1 when i
show a 2, and so on.

You may run into problems where your kernel is stuck for a
notebook is very slow and unresponsive, or your kernel loses
the following steps: 1. At the top of your screen, click **Kernel**, then **Intern**
help, click **Kernel**, then **Restart**. If you do this, you will have to run
of your notebook up until where you paused your work. 3. If tha
First, save your work by clicking **File** at the top left of your screen, **S**
Next, click **Control Panel** at the top right. Choose **Stop My Server** to s
My Server to start it back up. Then, navigate back to the noteb
still have to run your code cells again.

2.6 1.6. Submitting your work

All assignments in the course will be distributed as notebooks
your work from the notebook. We will use a system called OK
you submit. At the top of each assignment, you'll see a cell lik
to identify yourself. Run it and follow the instructions. Please
when logging in.

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When you finish an assignment, you need to submit it by runni
fine to submit multiple times. OK will only try to grade your fina
Don't forget to submit your lab assignment at the end of sec
everything.

[ ]:

3 2. Numbers

Quantitative information arises everywhere in data science. In a
to print out lines, expressions can represent numbers and me
expression `3.2500` evaluates to the number 3.25. (Run the cell an

[19]: 3.2500

[19]: 3.25

Notice that we didn't have to `print`. When you run a notebook cell, if
Jupyter helpfully prints out that value for you. However, it won't

[20]:
```
print(2)
```

```
3
4
```

2

[20]: 4

[23]: 4+(6*5-(6*3)) **2*((2**3)/4*7)

[23]: 2020.0

In standard math notation, the first expression is

$$3 + 6 \times 5 - 6 \times 3 \times \frac{2^3}{4} \times 7,$$

while the second expression is

$$2 \qquad (2^3)$$

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```
↪run it.
# You don't need to change this.
seconds_in_a_decade
```

[48]: 315532800

8

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